Prime-Power Divisibility at Lucas Entry Points

A complete explanation of the formula stated in OEIS A266587

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Abstract

OEIS A266587 describes the least index r of a Lucas number divisible by a given prime q, and labels a prime-power divisibility formula as a conjecture. We simplify its index expression and prove the stronger exact valuation formula $v_q(L_{rt}) = v_q(L_r) + v_q(t)$ for every odd t, whenever q is odd and $q \mid L_r$. Thus the displayed OEIS divisibilities follow for all parameters. The lifting property is classical; in particular, it is supported by Corollary 2.3 of Bugeaud–Luca–Mignotte–Siksek (2008). This note offers a self-contained derivation and does not claim novelty.

The OEIS assertion

Let $L_0 = 2$, $L_1 = 1$, and $L_{n+2} = L_{n+1} + L_n$. For an odd prime q which divides at least one positive-index Lucas number, let r be its smallest positive divisor index (the value $a(n) > 0$ for $q = \text{prime}(n)$ in A266587). The entry asserts, for $m > 1$ and $i \geq 0$, that

$$q^m \;\Big|\; L_{\,r+r(q-1)\sum_{j=1}^{m-1} q^{j-1}+2irq^{m-1}}.$$

As $1 + (q-1)\sum_{j=1}^{m-1} q^{j-1} = q^{m-1}$, the displayed index is simply $rq^{m-1}(2i+1)$.

Theorem 1. *Let q be an odd prime and $r \geq 1$ satisfy $q \mid L_r$. For every positive odd integer t,*

$$\boxed{v_q(L_{rt}) = v_q(L_r) + v_q(t).}$$

Consequently, for every $m \geq 1$ and $i \geq 0$,

$$v_q\big(L_{rq^{m-1}(2i+1)}\big) = v_q(L_r) + m - 1 + v_q(2i+1) \geq m,$$

proving the divisibility formula recorded in A266587.

An elementary lifting lemma

Lemma 1. *If q is an odd prime, $r \geq 1$, and $q \mid L_r$, then*

$$v_q(L_{ru}) = v_q(L_r) \quad (u \geq 1 \text{ odd},\ q \nmid u), \qquad v_q(L_{rq}) = v_q(L_r) + 1.$$

Proof. Let $\alpha = (1+\sqrt{5})/2$, $\beta = (1-\sqrt{5})/2$. Their powers satisfy $L_r = \alpha^r + \beta^r$ and $\alpha^r\beta^r = (-1)^r$. For odd t, the symmetric polynomial $X^t + Y^t$ is divisible by $S = X + Y$; its quotient is an integer polynomial $Q_t(S, B)$ in $S = X + Y$, $B = XY$ (by the symmetric-polynomial recurrence). It is even in S and has constant term

$$Q_t(0, B) = t(-B)^{(t-1)/2}. \tag{1}$$

Indeed, substituting $Y = -X$ and dividing the first-order vanishing by $X + Y$ gives $tX^{t-1} = t(-B)^{(t-1)/2}$. Therefore, with $S = L_r$ and $B = (-1)^r$, we have $L_{rt} = L_r Q_t(S, B)$.

If $q \nmid u$ and u is odd, (1), evenness in S, and $q \mid S$ show

$$Q_u(S,B) \equiv u(-B)^{(u-1)/2} \not\equiv 0 \pmod{q}.$$

This proves the first equality.

For $t = q$, the freshman's-dream congruence $(X+Y)^q \equiv X^q + Y^q \pmod{q}$ implies the polynomial congruence $Q_q(S,B) \equiv S^{q-1} \pmod{q}$. Hence

$$Q_q(S,B) = S^{q-1} + qR(S,B), \qquad R(0,B) = (-B)^{(q-1)/2}$$

for some integer polynomial R. As $q \mid S$, $q \geq 3$, and $B = \pm 1$,

$$Q_q(S,B) \equiv q(-B)^{(q-1)/2} \not\equiv 0 \pmod{q^2}.$$

Thus $v_q(Q_q(S,B)) = 1$, proving the second equality. □

Proof of Theorem 1. Write $t = q^h u$ with $h = v_q(t)$ and u positive, odd, and coprime to q. Apply the second formula of Lemma 1 successively h times to the indices $r, rq, \ldots, rq^{h-1}$; then apply the first formula with odd multiplier u. This gives exactly $v_q(L_{rt}) = v_q(L_r) + h$. □

Verification, historical status, and limitations

For $q = 3$ and $r = 2$, we have $L_2 = 3$ and $L_6 = 18$, so $v_3(L_6) = 2 = v_3(L_2) + 1$. For $q = 29$, $r = 7$ and $L_7 = 29$, the theorem predicts $29^m \mid L_{7 \cdot 29^{m-1}(2i+1)}$. Exact arithmetic checks verified the stated valuation for sampled odd primes $q < 101$, odd multipliers, and indices below 2000; these checks are not part of the proof.

The OEIS expression is an instance of *established* Lucas divisibility theory, not a newly resolved independent open problem: see the 2008 paper cited below, especially its Corollary 2.3. We establish only this stated divisibility/valuation assertion, not other conjectures concerning which primes divide Lucas numbers.

References and disclosure

1. OEIS Foundation, *A266587: Least Lucas-number index divisible by the n-th prime*, https://oeis.org/A266587 (accessed 2026-11-14).
2. Y. Bugeaud, F. Luca, M. Mignotte, and S. Siksek, "Almost powers in the Lucas sequence," *Journal de Théorie des Nombres de Bordeaux* **20** (2008), 555–600; see Corollary 2.3, p. 561. https://www.numdam.org/article/JTNB_2008__20_3_555_0.pdf.

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